
Dinámica Neuronal

72.25 Simulación de Sistemas

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Introducción

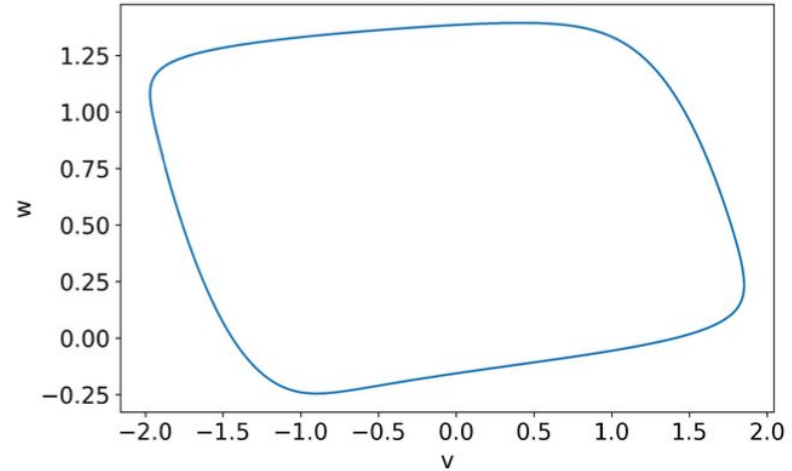
Modelo FitzHugh-Nagumo

Potencial de membrana:

$$\frac{dv_i}{dt} = v_i - \frac{v_i^3}{3} - w_i + I + K \sum_j A_{ij} (v_j - v_i)$$

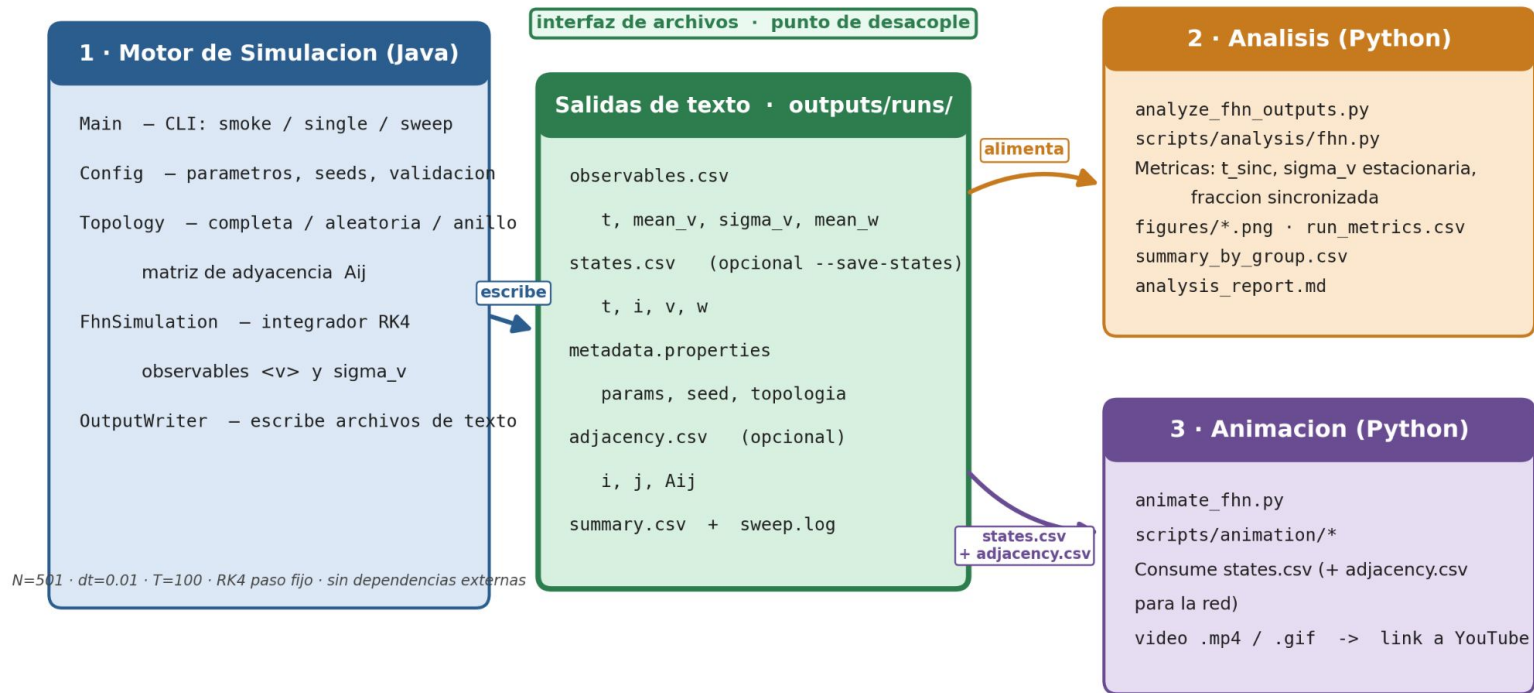
Variable de recuperación:

$$\frac{dw_i}{dt} = \epsilon (v_i + a - bw_i)$$



Implementación

Arquitectura



Pseudocódigo

```
SimularFHN(config, topología) <- observables
  N, K, dt, T, Δsave <- config
  rng <- Random(config.runSeed)
  para i en [0..N-1]:
    v[i] <- U(-0.05, 0.05); w[i] <- U(-0.05, 0.05)

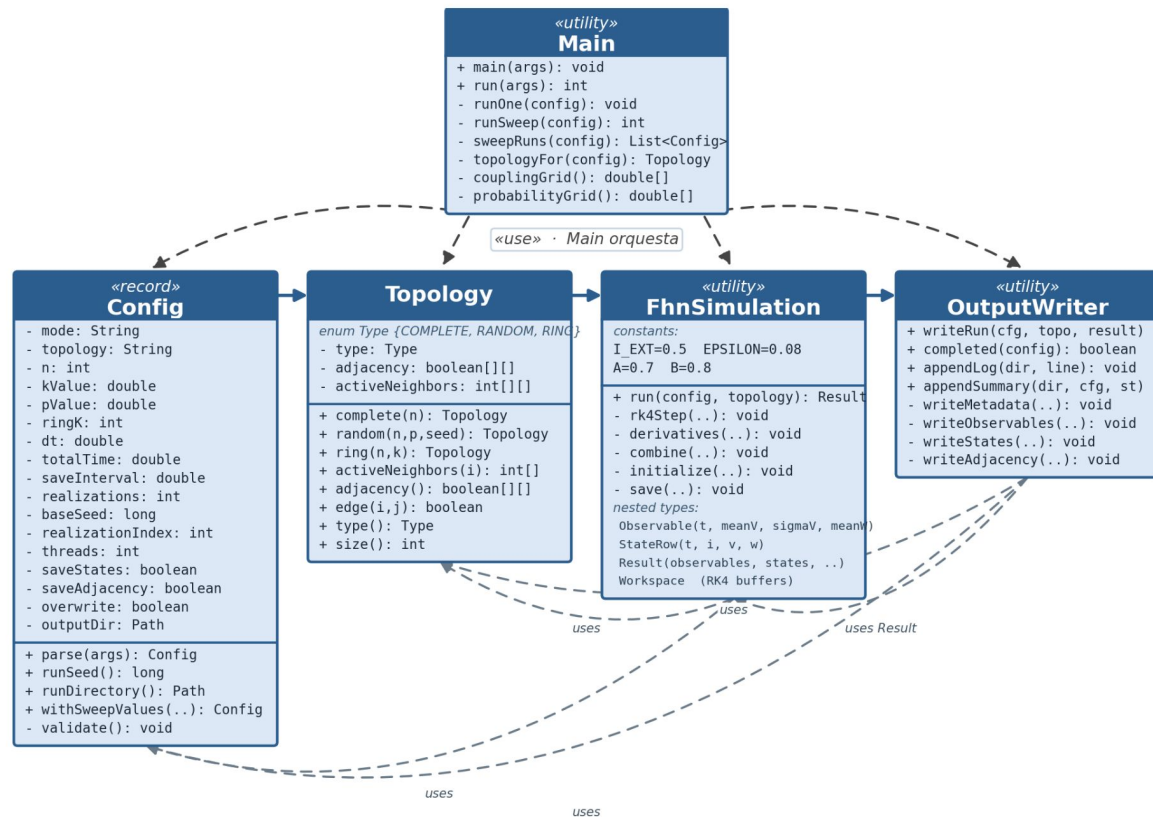
  t <- 0; próximoT <- Δsave; Guardar(t, v, w)
  mientras t < T:
    paso <- min(dt, T - t)
    (v, w) <- PasoRK4(v, w, topología, K, paso)
    t <- t + paso
    si t >= próximoT o t >= T: Guardar(t, v, w); próximoT <- próximoT + Δsave
  devolver observables

PasoRK4(v, w, topología, K, dt):

Derivadas(v, w, topología, K):

Guardar(t, v, w):
   $\langle v \rangle$  <- media(v);  $\sigma_v$  <- sqrt(media((v -  $\langle v \rangle$ )2));  $\langle w \rangle$  <- media(w)
```

Diagrama UML



Simulaciones

Parámetros de entrada

Parámetros de entrada variables

v_i	$[-0.5, 0.5]$
w_i	$[-0.5, 0.5]$

Parámetros de entrada fijos

I	0.5
ϵ	0.08
a	0.7
b	0.8

Observables

Potencial promedio de la red:

$$\langle v(t) \rangle = \frac{1}{N} \sum_i v_i(t)$$

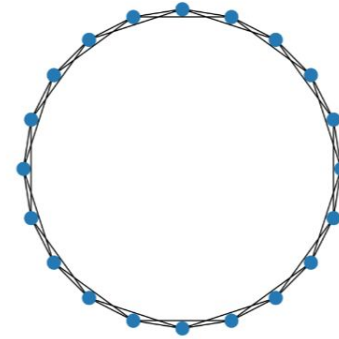
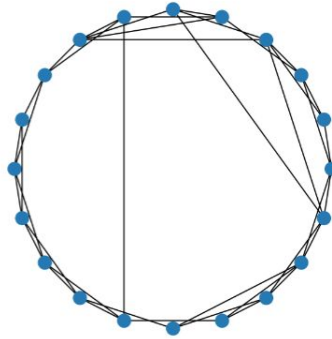
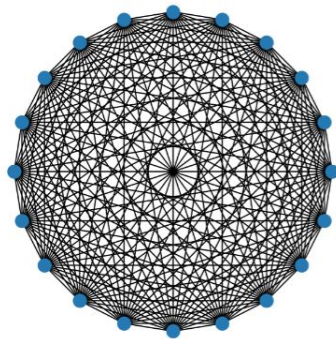
Dispersión espacial:

$$\sigma_v(t) = \sqrt{\frac{1}{N} \sum_i (v_i(t) - \langle v(t) \rangle)^2}$$

Observaciones

Se estudian 3 redes distintas:

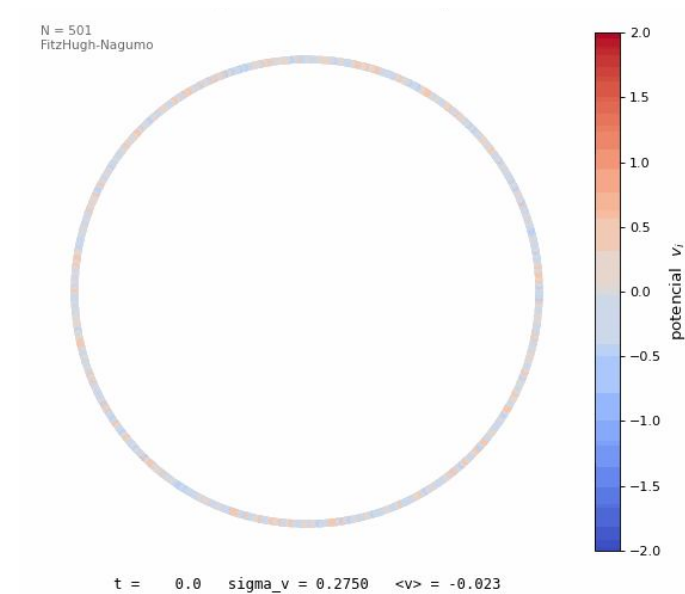
Red totalmente conectada	Red aleatoria	Red anillo
$A_{ij} = 1, \forall i \neq j$	$A_{ij} = 1, \text{ con } p = [0, 1]$	$A_{ij} = 1 \Leftrightarrow j = [i-k, \dots, i-1, i+1, \dots, i+k],$ con $k = [1, 10]$



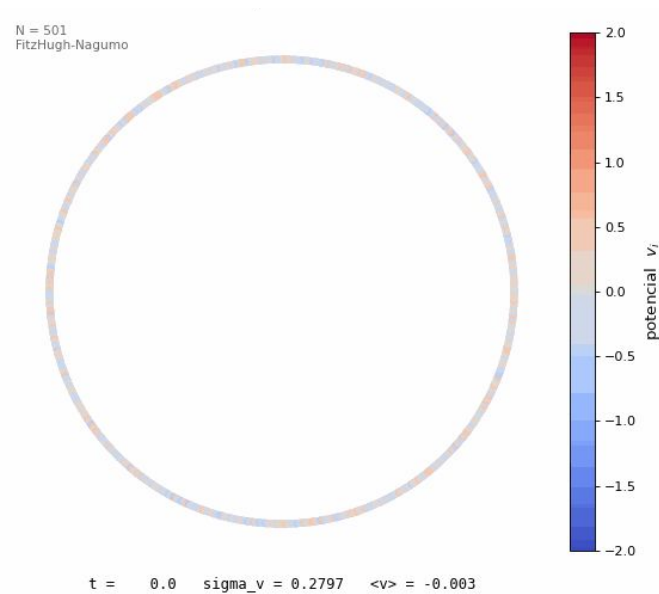
Resultados

Animaciones

$K = 0$

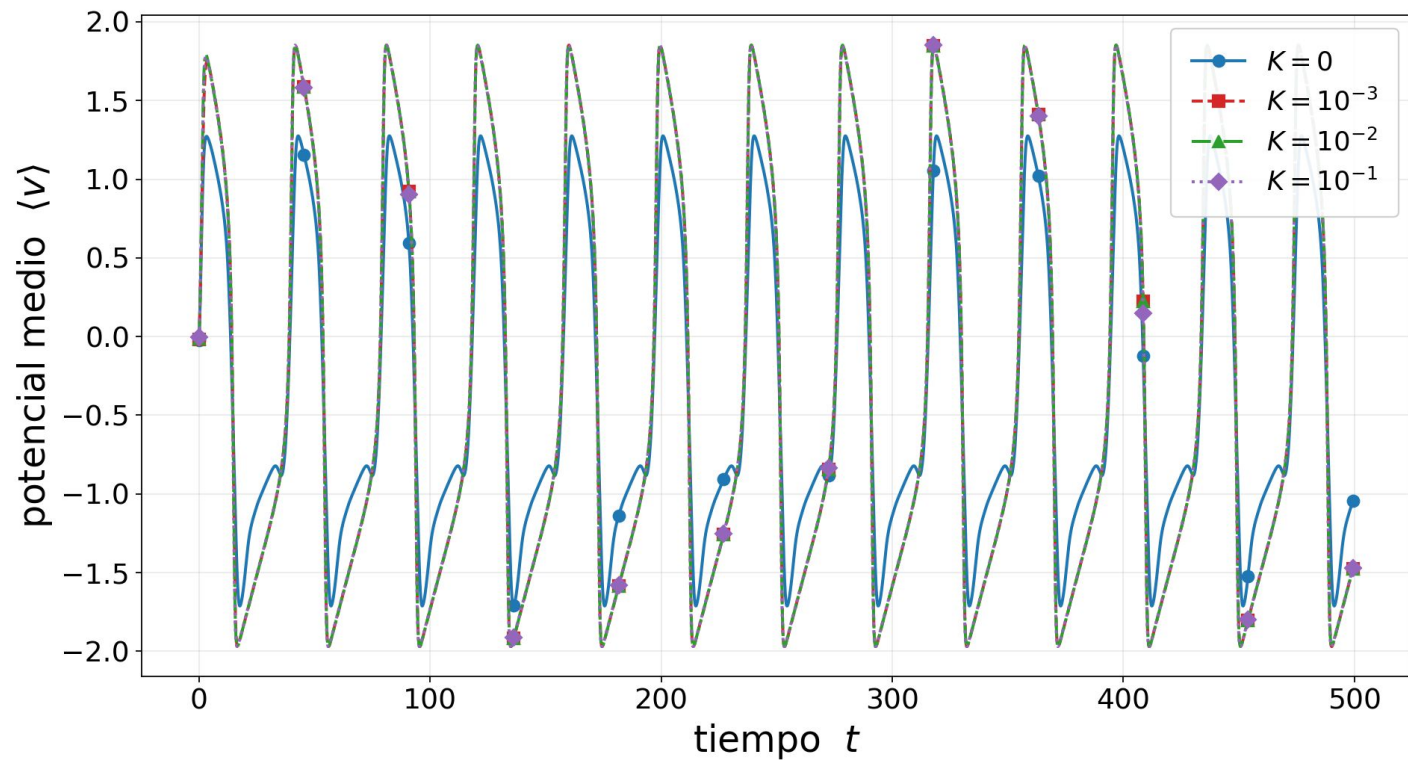


$K = 0.1$



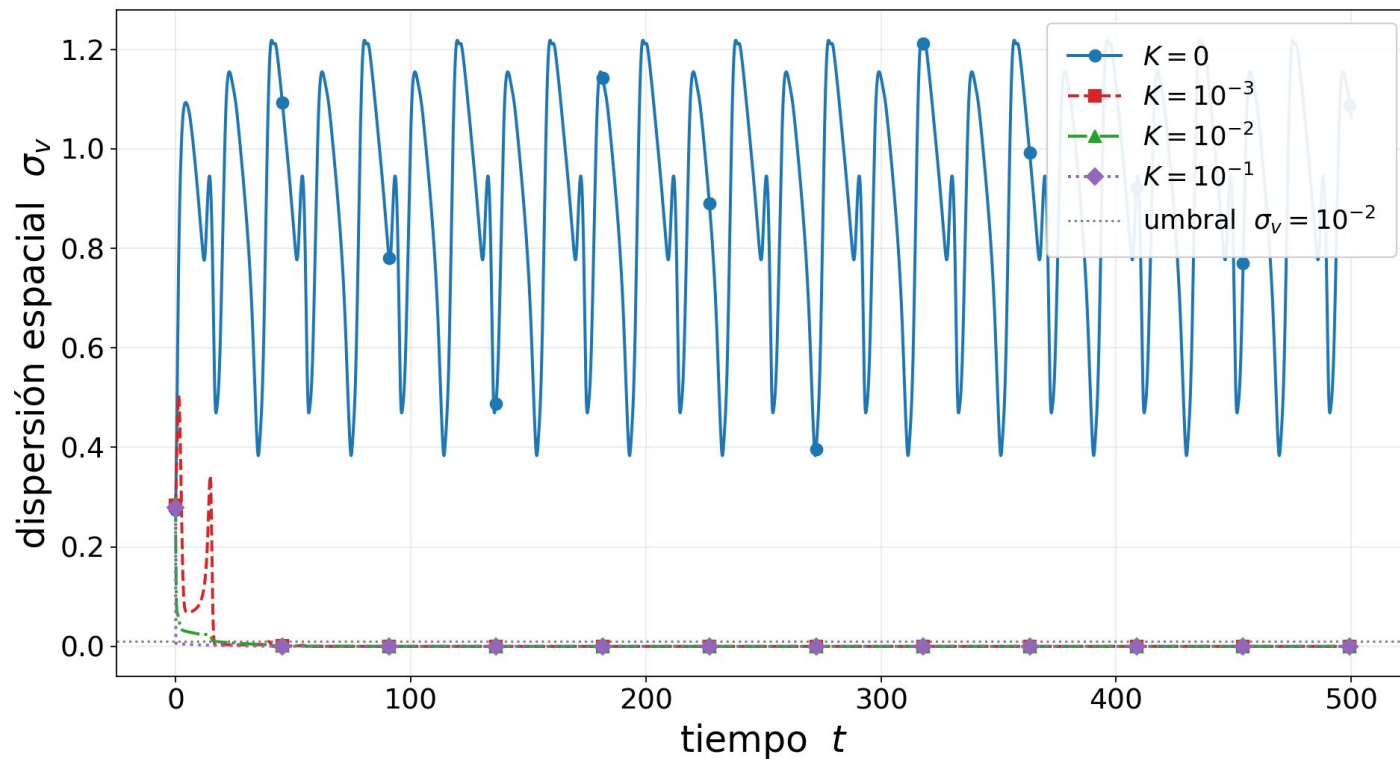
Resultados: Red totalmente conectada

Evolución temporal de $v(t)$



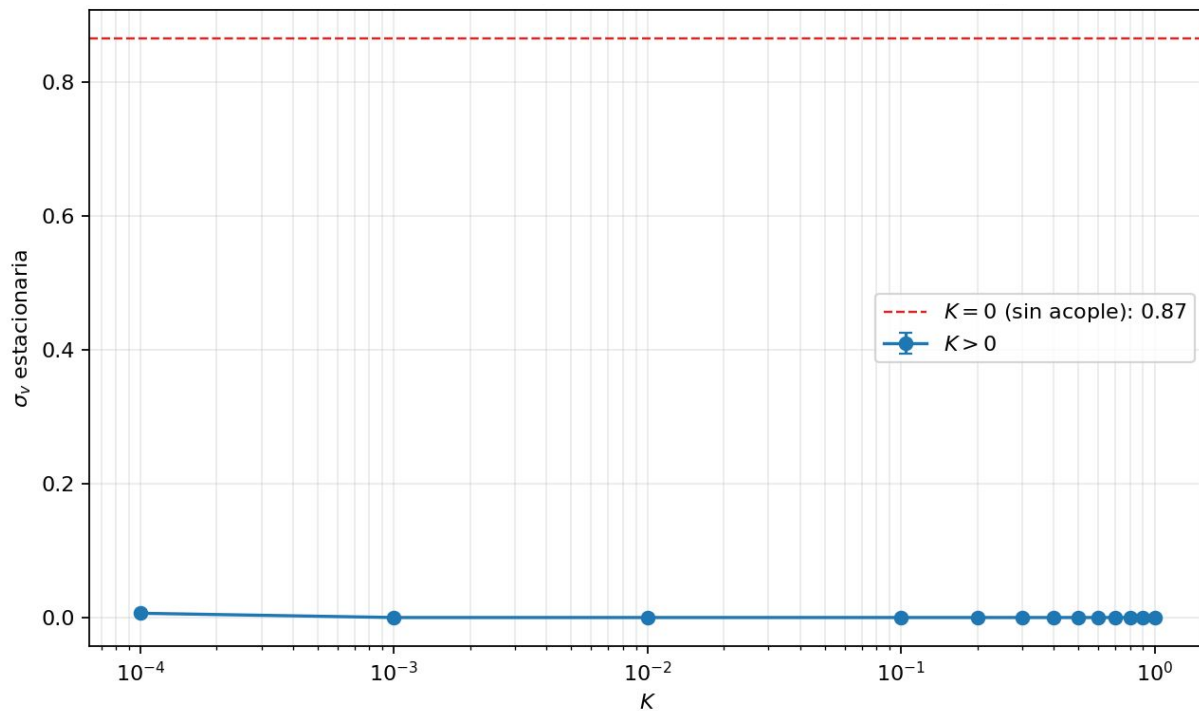
Resultados: Red totalmente conectada

Evolución temporal de $\sigma_v(t)$



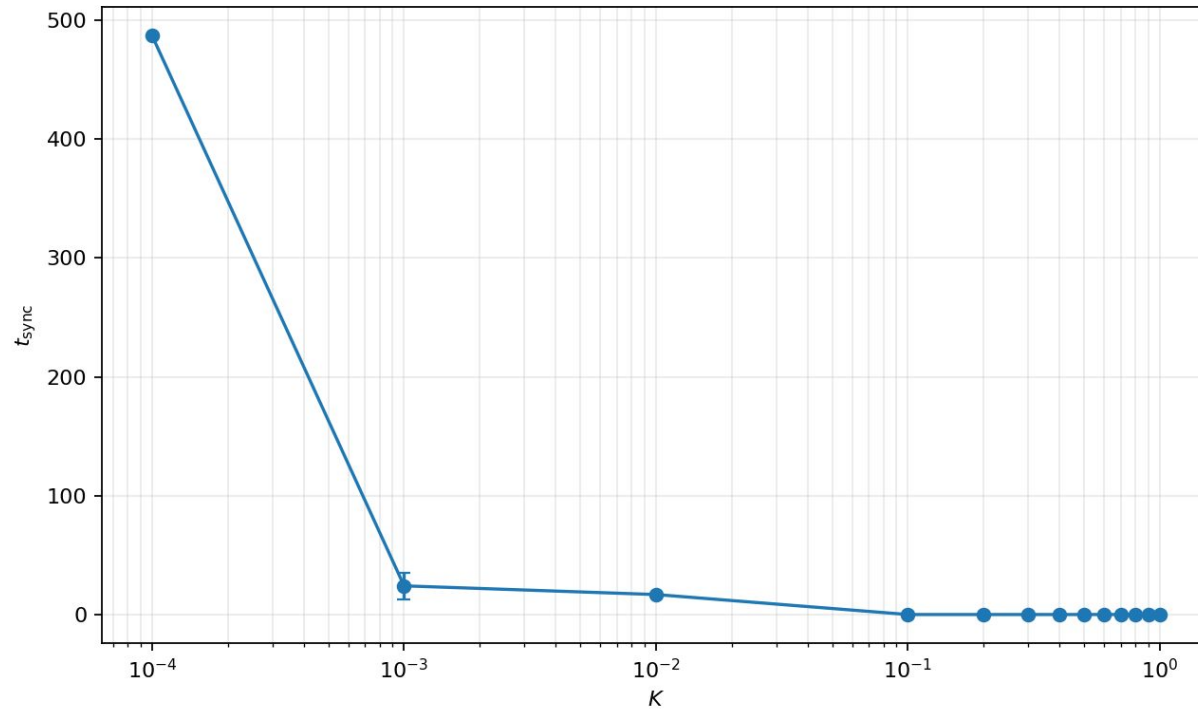
Resultados: Red totalmente conectada

Estado estacionario



Resultados: Red totalmente conectada

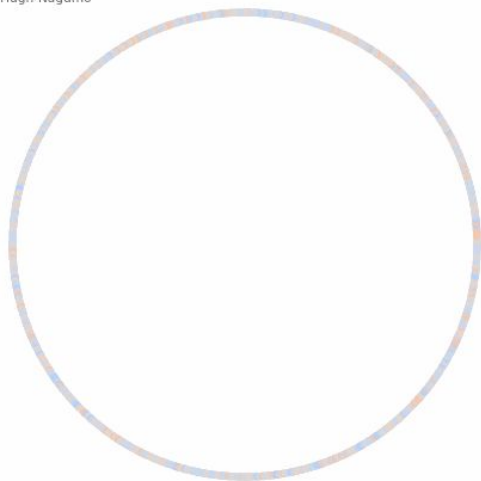
Sincronización de t y K



Animaciones

$$p = 0.01 - K = 0.1$$

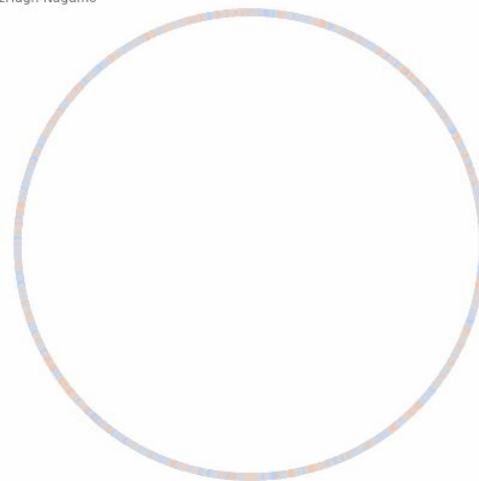
N = 501
FitzHugh-Nagumo



t = 0.0 sigma_v = 0.2804 <v> = -0.025

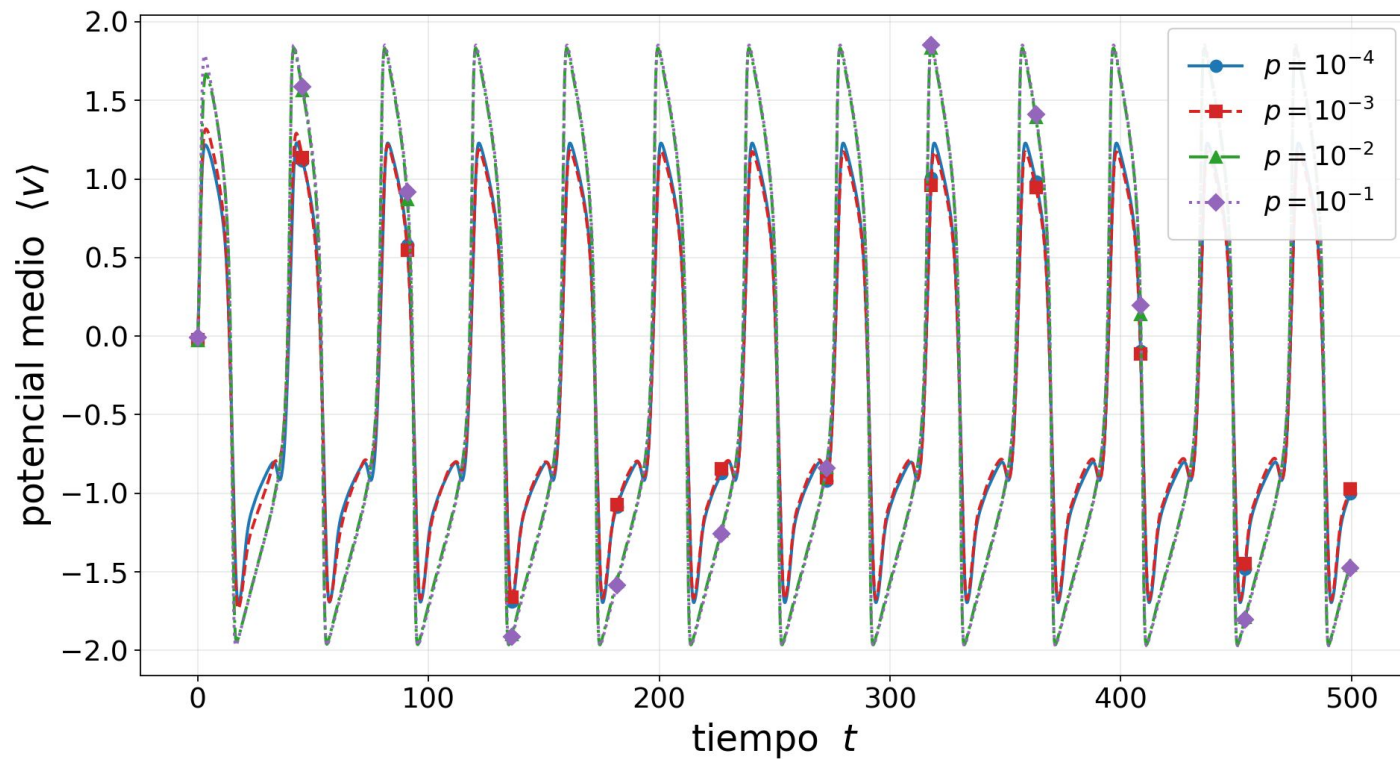
$$p = 0.1 - K = 0.1$$

N = 501
FitzHugh-Nagumo

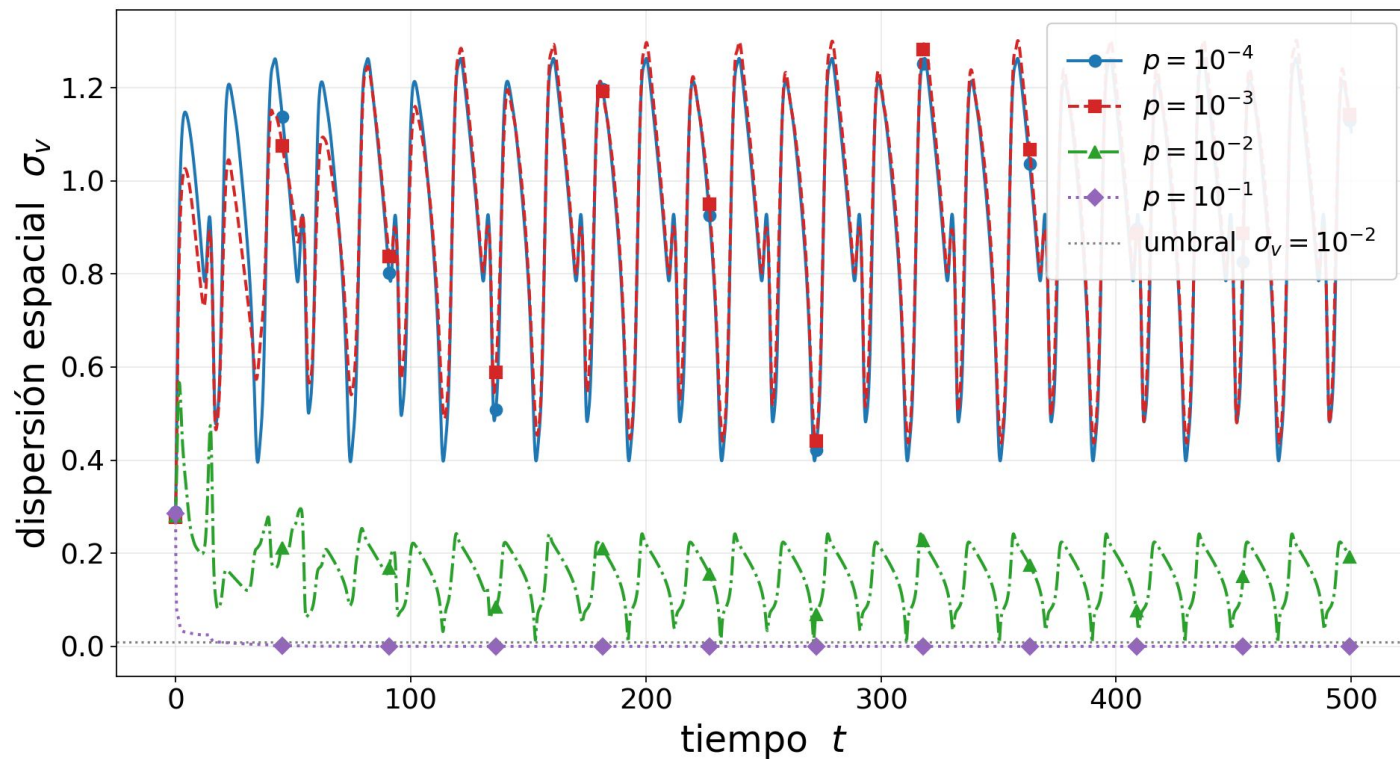


t = 0.0 sigma_v = 0.2855 <v> = -0.010

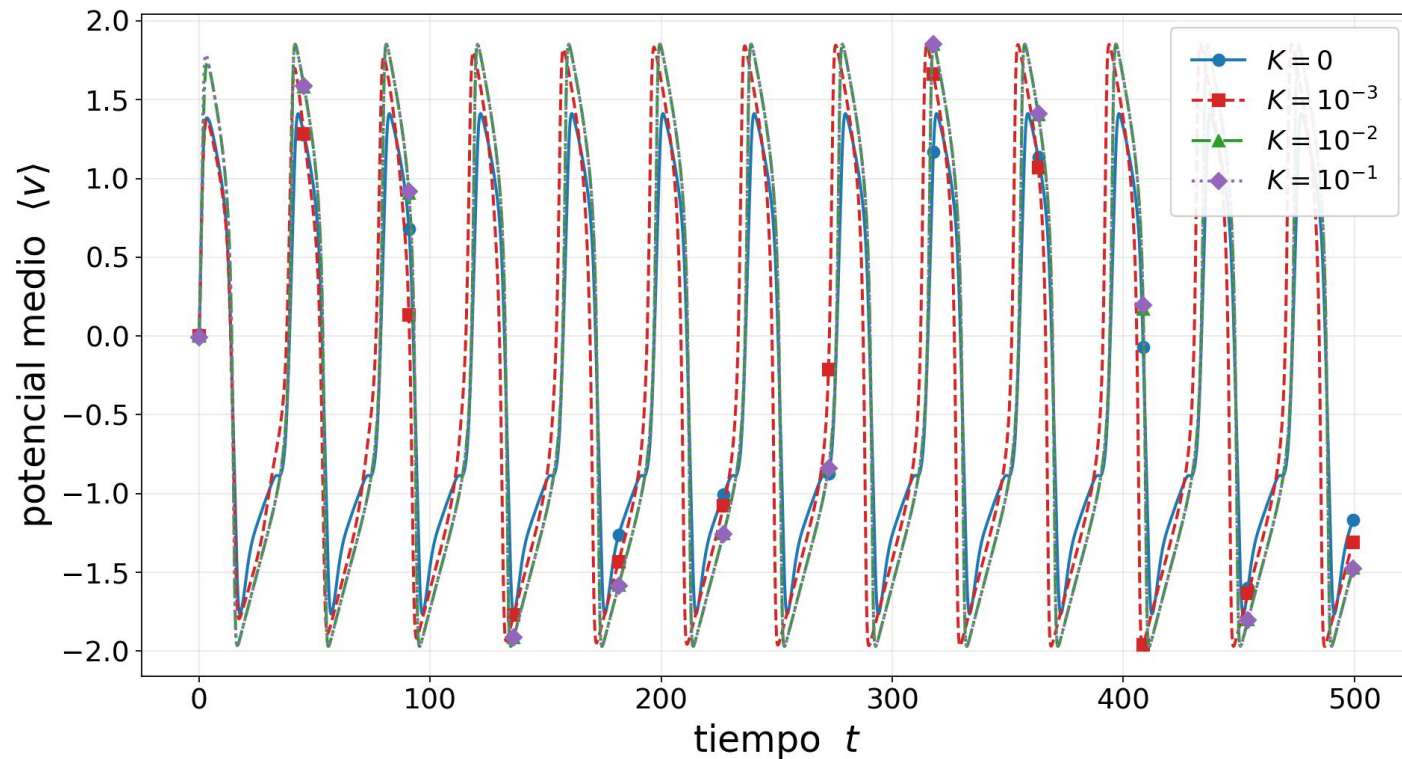
Evolución temporal de $v(p)$



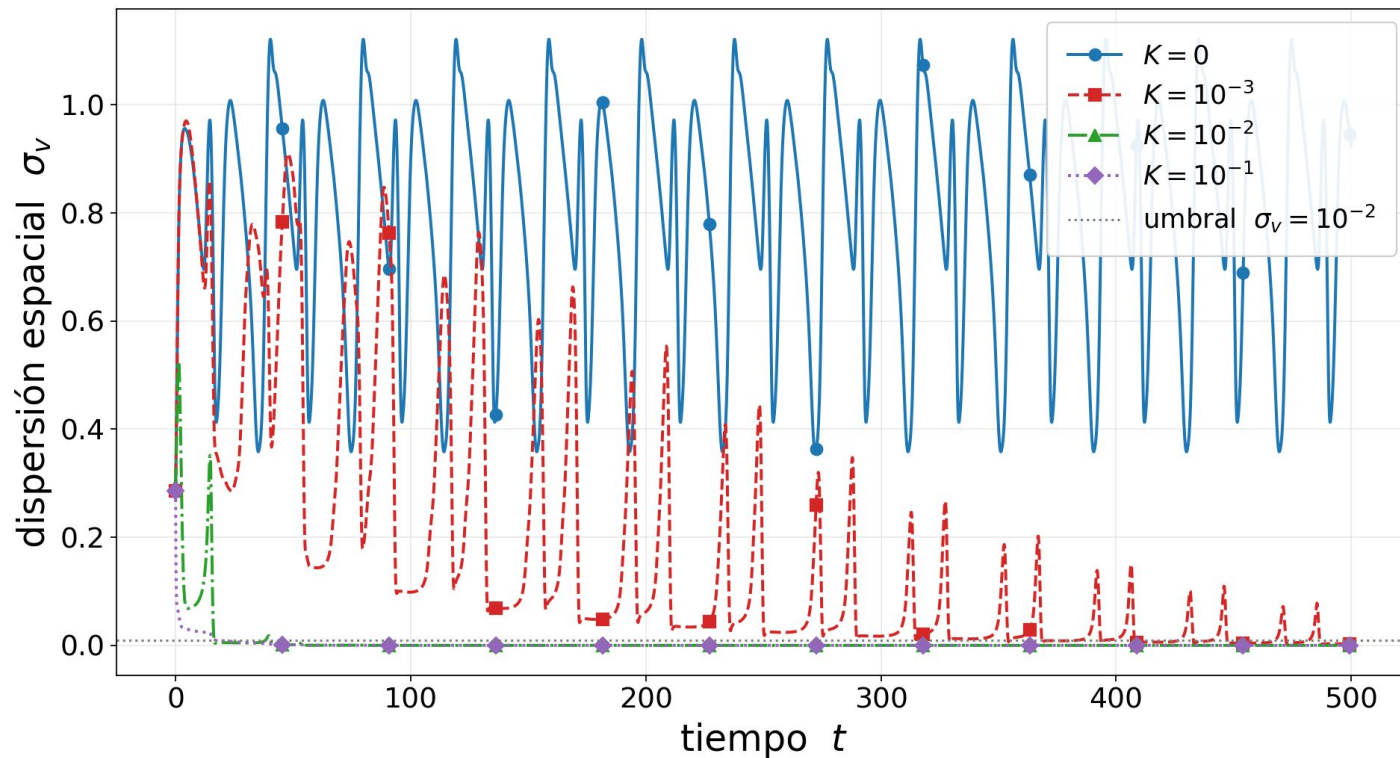
Evolución temporal de $\sigma_v(p)$



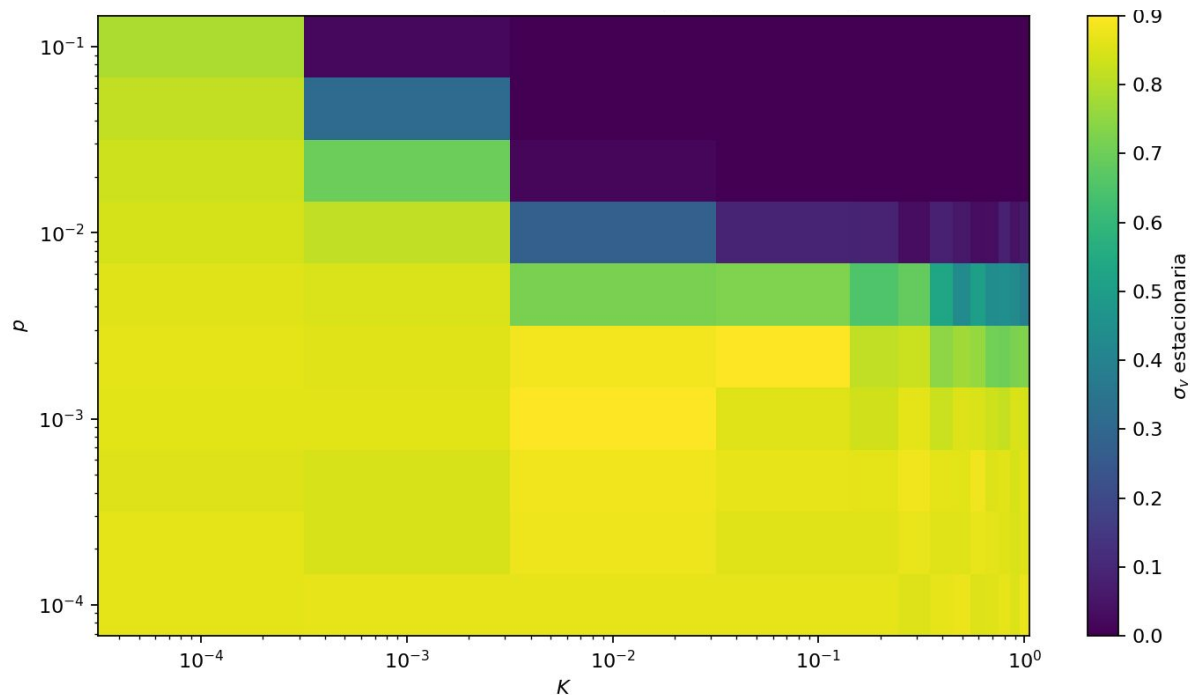
Comportamiento con distintos K



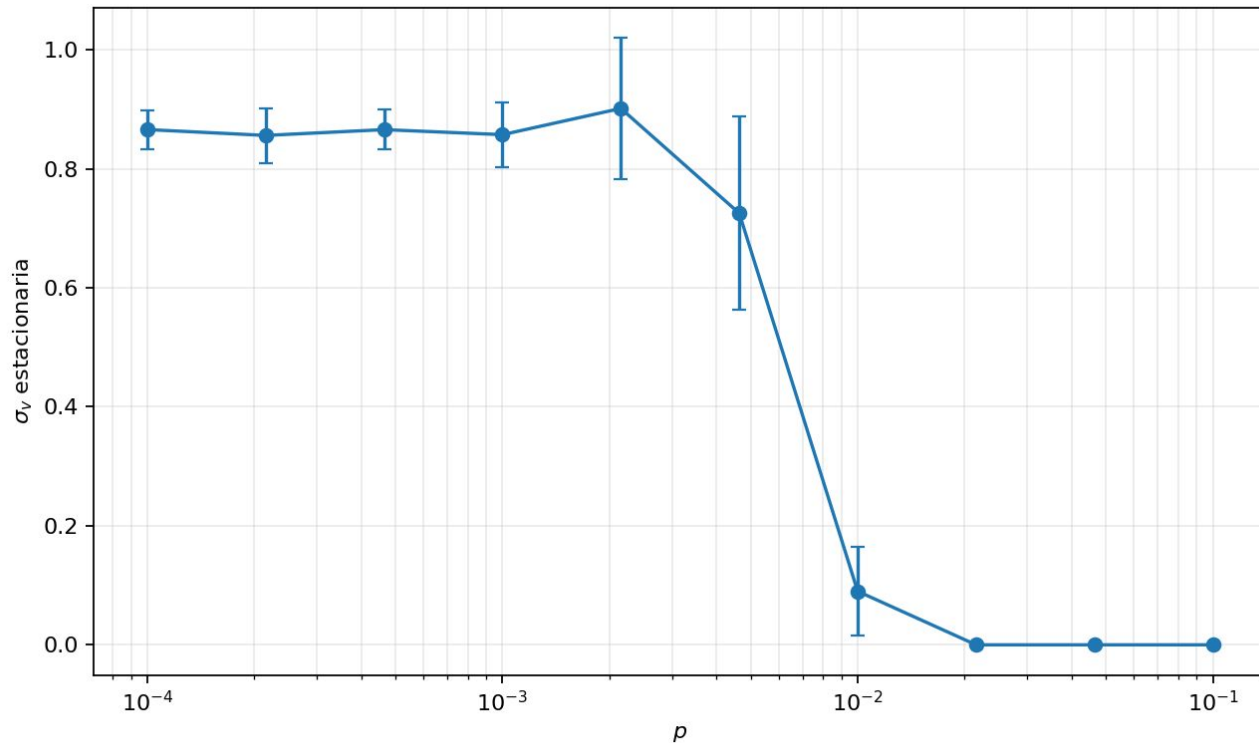
Comportamiento con distintos K



Dispersión en función de p y K



Sincronización de p y K

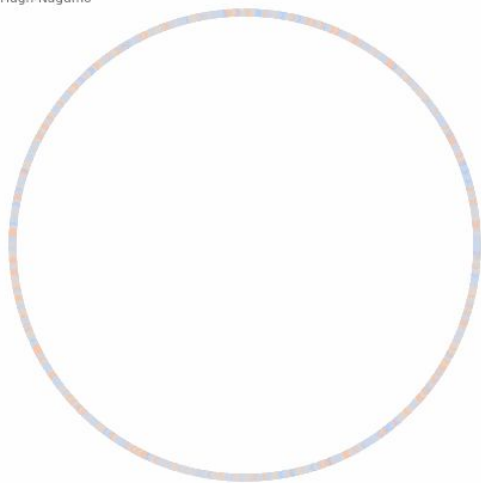


Animaciones

$$k = 1 - K = 0.1$$

Red anillo - $k=1$, $K=0.10$ (lento, no sincroniza)

N = 501
FitzHugh-Nagumo

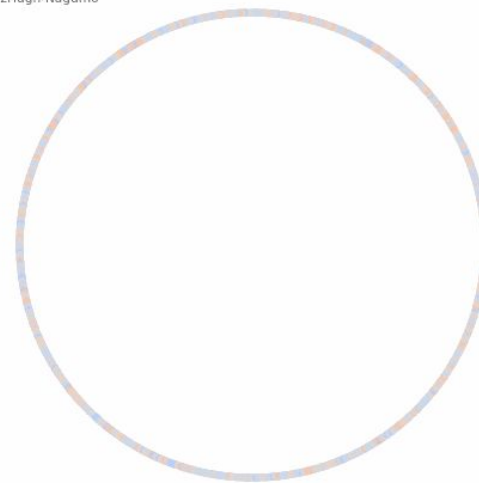


t = 0.0 sigma_v = 0.2965 <v> = +0.013

$$k = 10 - K = 0.1$$

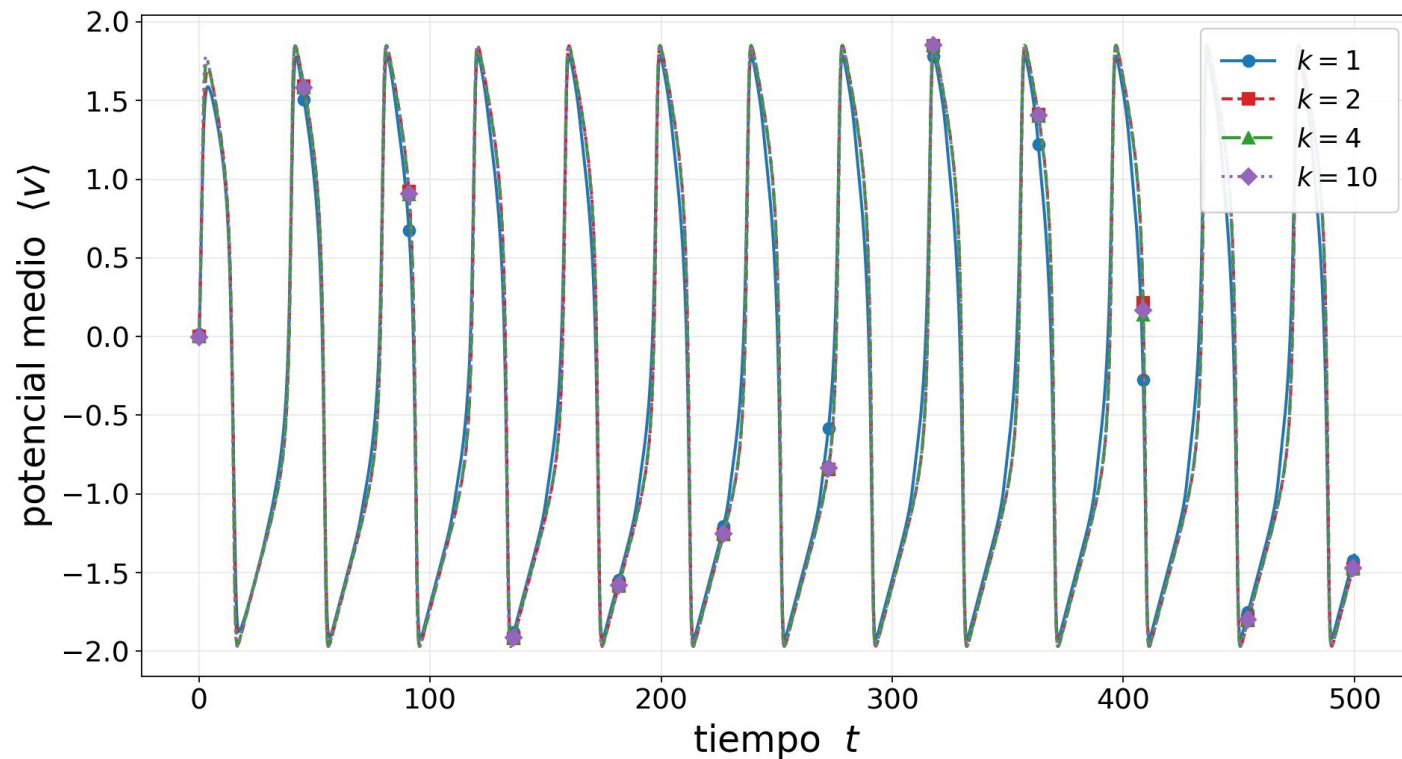
Red anillo - $k=10$, $K=0.10$ (rapido)

N = 501
FitzHugh-Nagumo

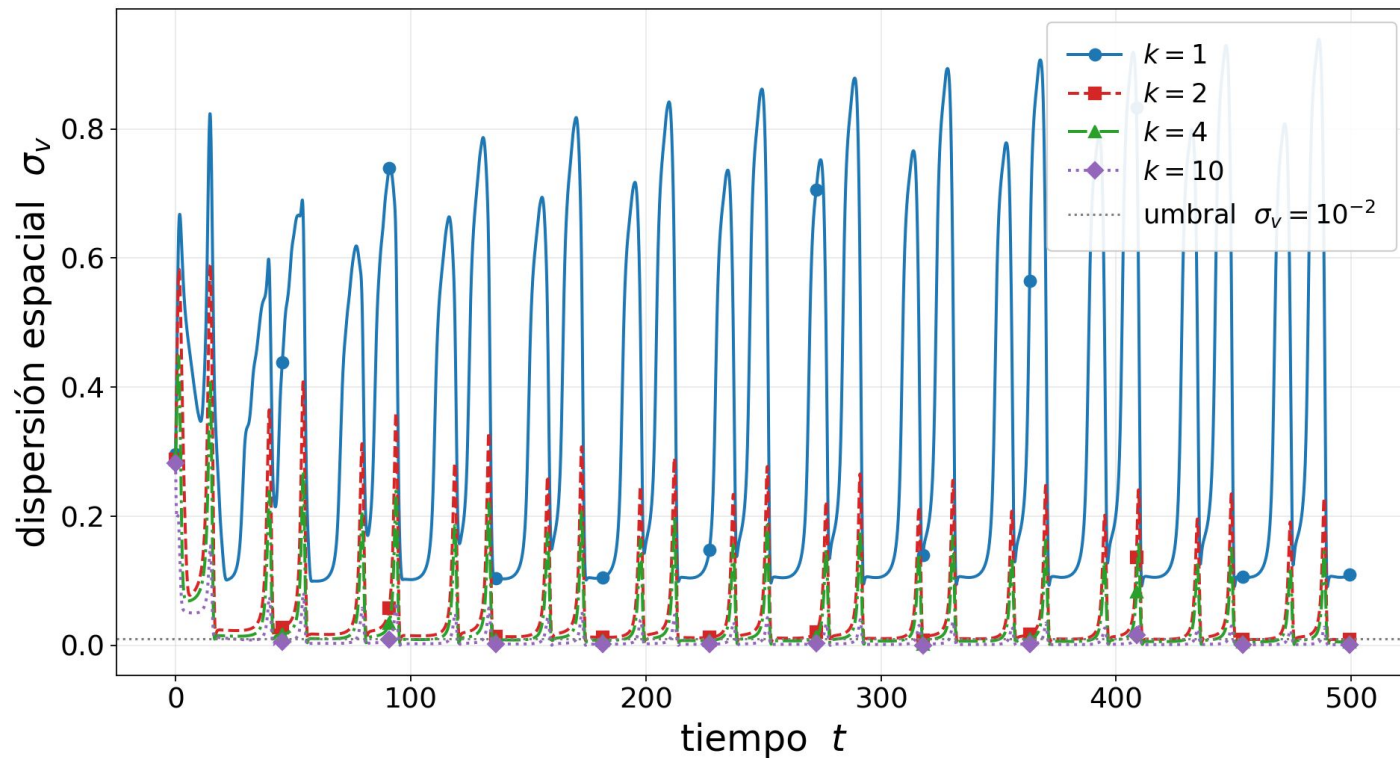


t = 0.0 sigma_v = 0.2818 <v> = -0.002

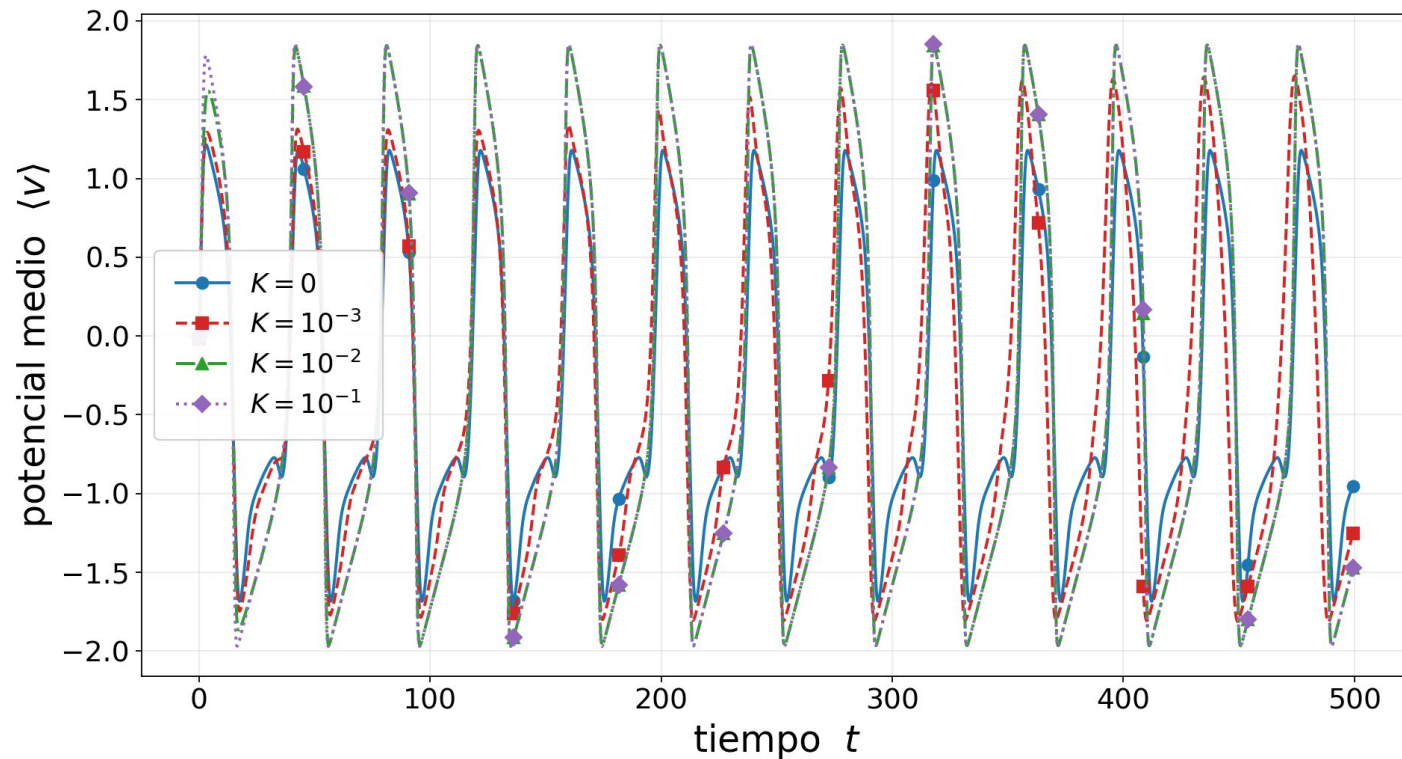
Evolución temporal de $v(k)$



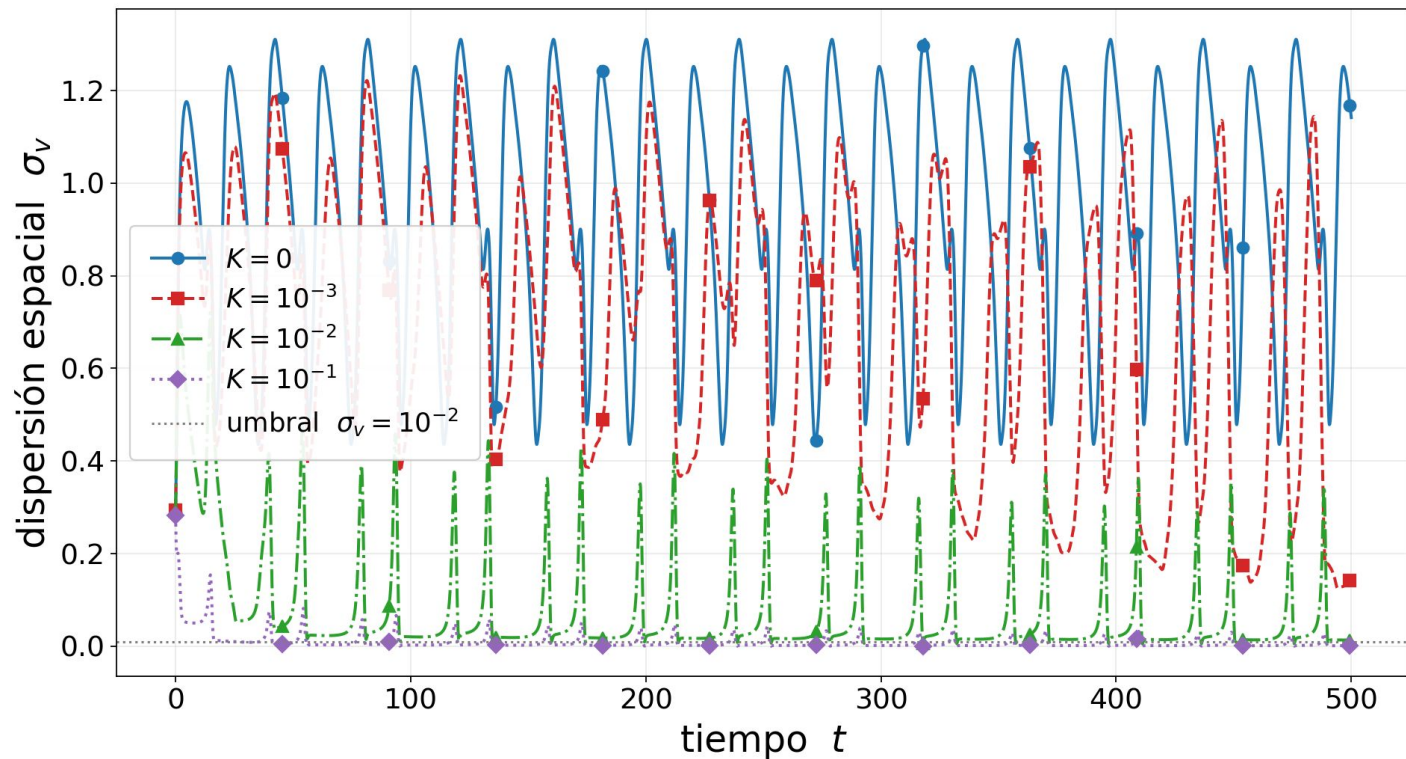
Evolución temporal de $\sigma_v(k)$



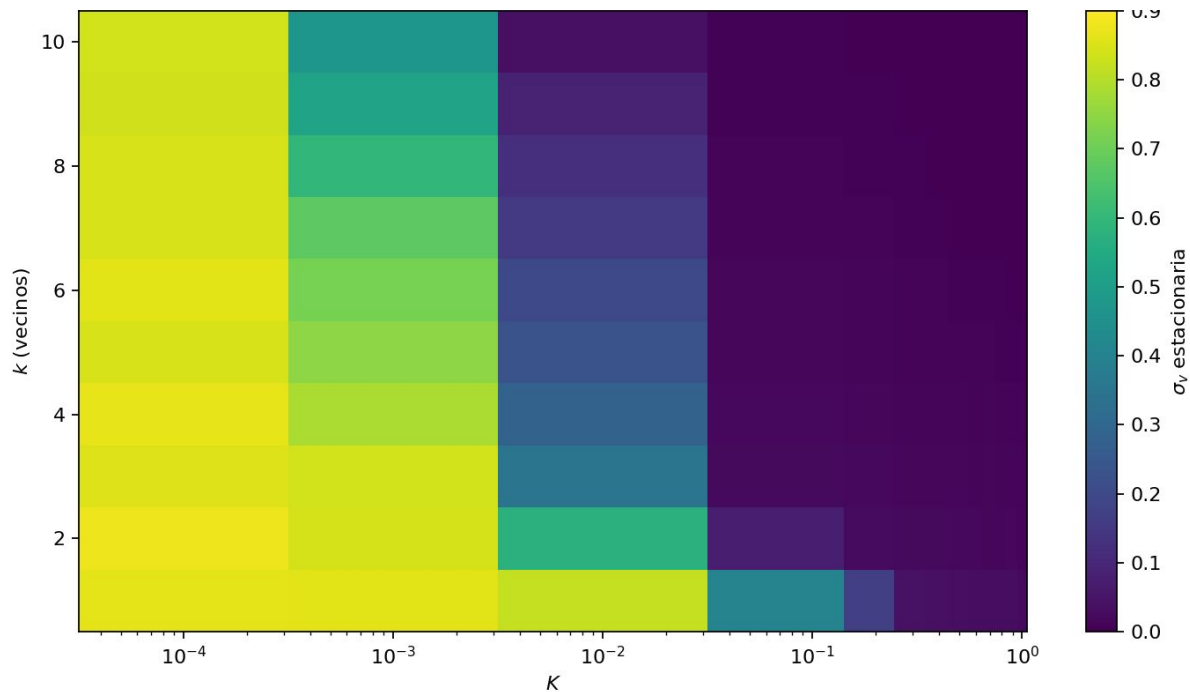
Comportamiento con distintos K



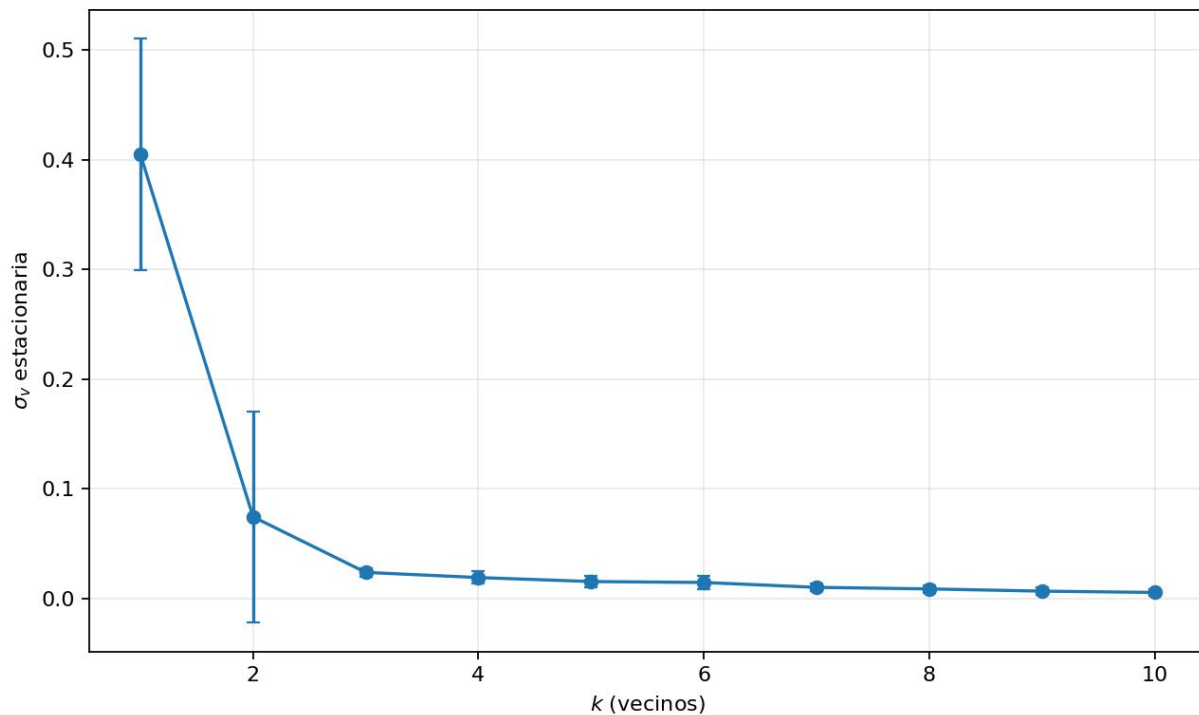
Comportamiento con distintos K



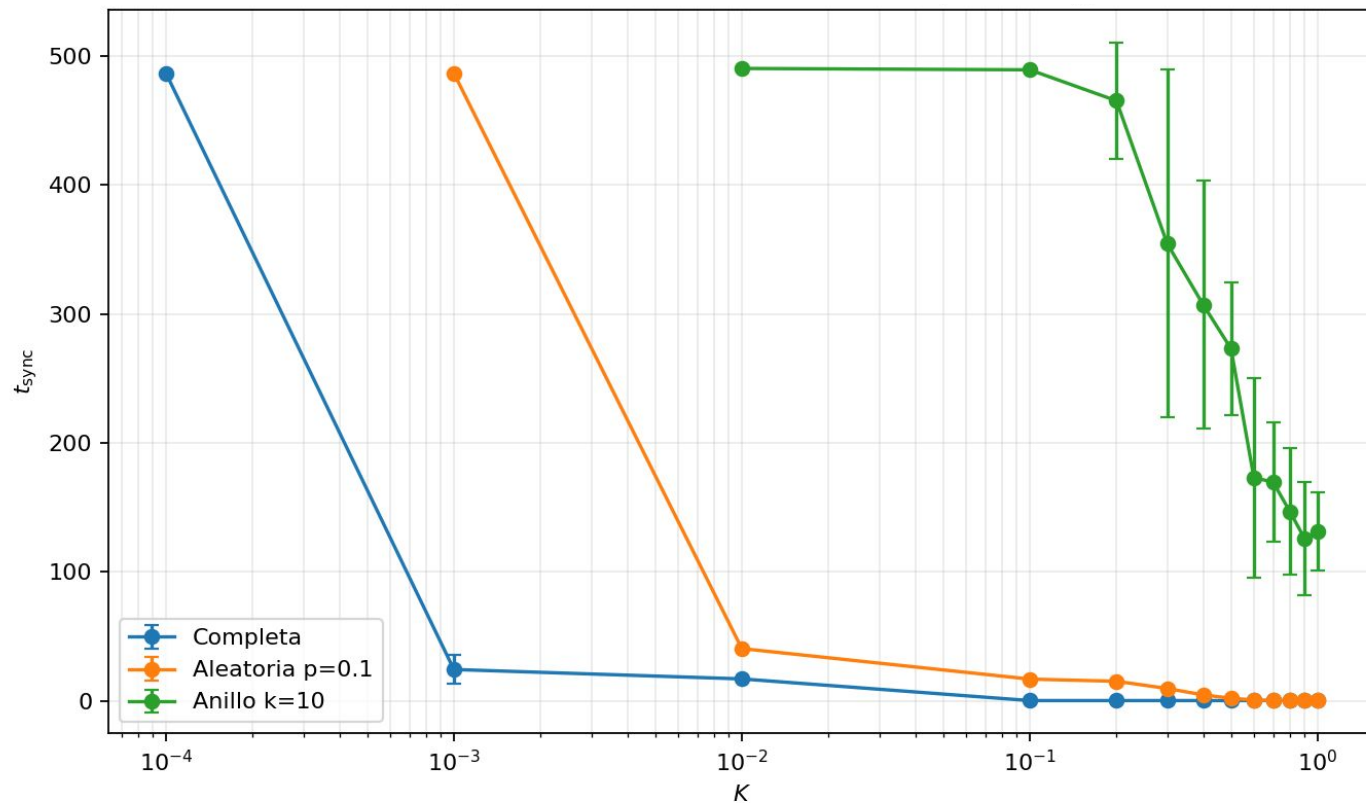
Dispersión en función de k y K



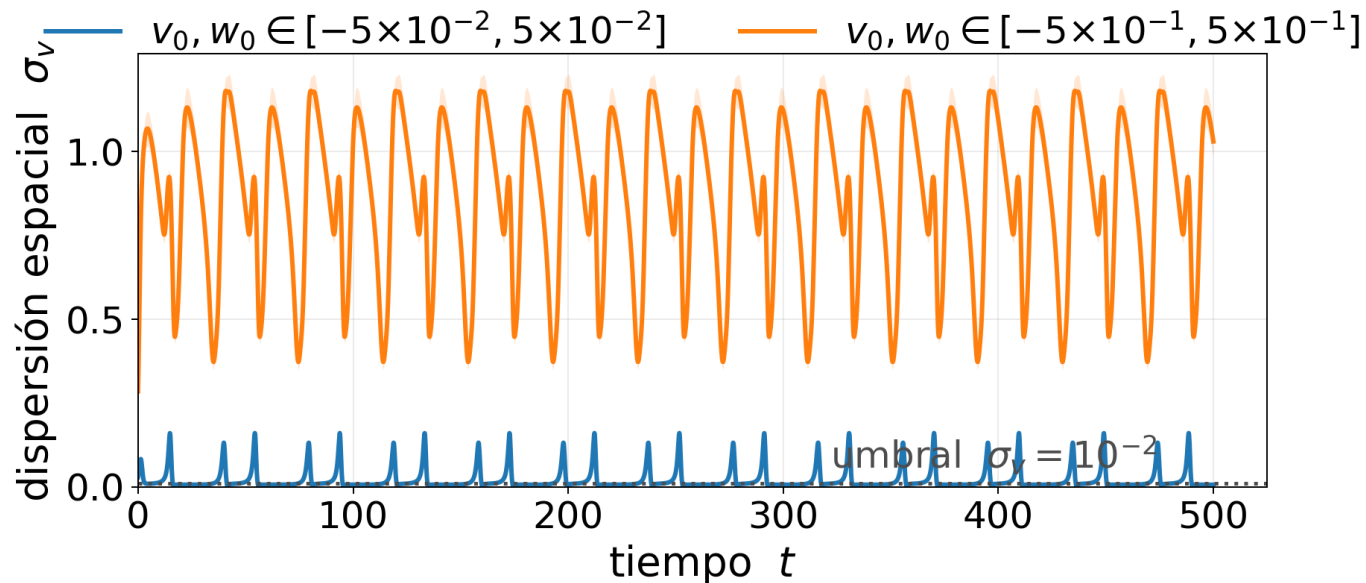
Sincronización de k y K



Tiempo al estacionario entre redes



Efecto de la condición inicial



Parámetros fijos

$N = 501$

$K = 0$

$T = 500$

15 realizaciones

Números clave

$\sigma_v(0)$:

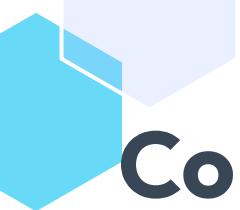
$2,9 \times 10^{-2}$ (angosta)

vs $2,9 \times 10^{-1}$ (ancha)

sincronizan:

15/15 vs 0/15

Conclusiones



Conclusiones

1. La sincronización tiene un umbral de acople y, lejos de ese umbral, aparece muy rápido.
2. El acople crítico no es único: depende fuertemente de la conectividad.
3. Cambiar las condiciones iniciales era necesario para no forzar una sincronización artificial.

Muchas gracias!

